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IMAGE COMPRESSION Using Discrete Cosine Transform Technique

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Presented here is a MATLAB-based program for image compression using discrete cosine transform technique. It works for both coloured and grayscale images.

Over the last few years, messaging apps like WhatsApp, Viber and Skype have become increasingly popular. These applications let users send and receive text messages and videos. All of us make extensive use of these applications without knowing what actually goes behind the scene in transmitting high-quality images and text. This article dwells on the image/video compression concept that is being used by nearly all the Internet-based messaging applications.

Fig. 1 shows screenshots of the author's mobile as it compresses the video using WhatsApp software. The application performs compression in two steps: (a) preparing mode and (b) sending mode. The sending mode basically deals with transmission of data stream onto the communication channel. So here we will restrict ourselves to discussing the typical image/video compression algorithm that runs behind the preparing mode.

The main job of the image/video compression algorithm is to reduce the size of the file to be transmitted. For example, in the case of a 5MB video file, the image/video compression software running behind the preparing mode in WhatsApp software makes the video smaller by up to 1MB, thus saving the memory space for transmission to take place. The software performs this process automatically as you send the file to one of your friends. However, this process is visible only when the file to be transmitted is large in size, such as the 5MB video in this example.

Image compression is the task of representing an image with minimum number of coefficients so that the total memory occupied by the compressed image is much less than the original image. With this reduction of memory requirement for high-definition image, the transmission of these images onto the transmitting medium is much easier than without compression.

In order to achieve the task of image compression, it has to be represented in a domain where high-definition images/videos are sparse. The two existing domains widely used in digital signal processing are spatial domain and frequency domain.

The third domain widely used nowadays in the field of image processing is sparse domain. In this domain, mostly the coefficients are sparse in nature, i.e., most of them are zero with very few non-zero coefficients. Typically used techniques for transforming the spatial domain to sparse domain include wavelet, curvelet, singular value decomposition (SVD) and discrete cosine

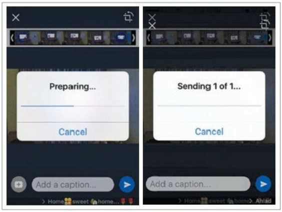

Fig. 1: Compression process in mobile using WhatsApp software: (a) Application in preparing mode, (b) Application in sending mode

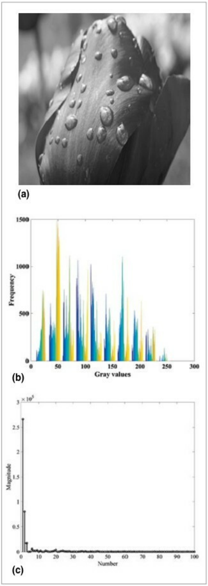

Fig. 2: (a) Original image, (b) Spatial domain (histogram) and (c) Sparse representation